Design of Self-Sensing Electrohydraulic Actuator Powered by Induced Interfacial Charges (EPIC) via Capacitance Sensing Circuit

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***Abstract*—This paper proposes a design of self-sensing electrohydraulic actuator powered by induced interfacial charges (EPIC) via capacitance sensing circuit. EPIC was designed to be a powerful soft actuator with high performance for output force and low input voltage. We improved it to a smart actuator by adding self-sensing feature to extend its application as a smart haptic device that allows interactive operations based on external haptic stimuli and feedback-controlled varifocal lens. The self-sensing was accomplished by sensing the capacitance variation of the actuator, where a mixed-signal-based self-sensing method is adopted. Prior to designing the self-sensing circuit, electrochemical impedance spectroscopy was conducted to analyze the electrical characteristics of EPIC. From the result, the equivalent circuit of EPIC under DC-biased operation condition was modeled as a parallelly connected resistor and capacitor. In addition, the relationship between the contact electrode area and capacitance of the EPIC actuator was found to be proportional. Based on the electrical characteristics of EPIC, we designed a sensing circuit to measure contact force in accordance with the capacitance variation of the actuator.**

I. Introduction

Recently, soft actuators have received significant attention as a new technology that enable safe human-robot interaction, which is not possible using traditional rigid actuators [1]. However, most of the present soft actuators control their output through the open-loop systems or using external sensors that make inaccurate and bulky. Therefore, many researchers are studying novel self-sensing soft actuators that can sense their own displacements or forces [1], [2]. Self-sensing soft actuators have many advantages, such as more accurate outputs produced through closed-loop control and sensing the external environment without additional sensors [3], [4]. For instance, dielectric elastomer actuators (DEAs) and Peano-HASEL (hydraulically amplified self-healing electrostatic) actuators were demonstrated as controllable robotic arms (errors < 4%) [3] and haptic feedback devices [4] with self-sensing capabilities.

Our research group has developed a novel soft actuator called an electrohydraulic actuator powered by induced interfacial charges (EPIC), in which the operating voltage is reduced and output force is amplified compared with the conventional dielectric electroactive polymer (EAP) actuator [5]. This actuator has been demonstrated as a powerful artificial muscle (output force > 4 N, which is 16 times that of the actuator weight), wide-bandwidth vibrotactile haptic device (maximum 6 G acceleration performance under 90–700 Hz), and high-resolution varifocal lens (focal length variation > 350%) [5], [6]. If the EPIC actuator can measure its own displacement or force by self-sensing, then the actuator can have extended and improved applications beyond its present capabilities. Examples of such applications include a feedback-controlled EPIC actuator, which is especially beneficial for controlling the focal length of the EPIC varifocal lens, and a smart haptic device with an interactive operation to external stimuli, such as touch and pressure.

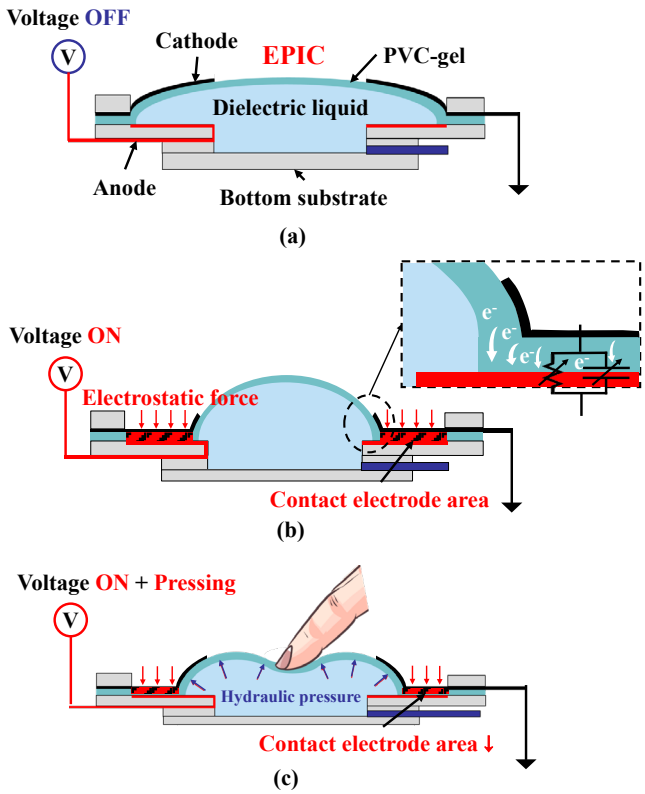


Figure 1. (a) Configurations of EPIC actuator (under voltage off), (b) morphed state by zipping under electrostatic force due to HSCD (under voltage on), and (c) decreasing contact electrode area by dielectric liquid penetration (under voltage on and pressing).

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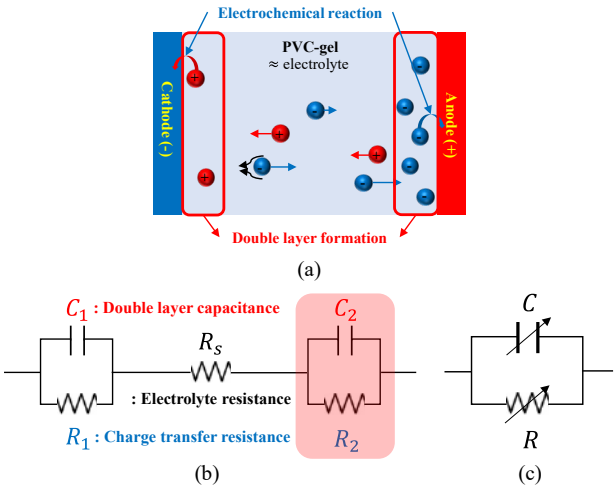


Figure 2. Electrical analysis of EPIC actuator: (a) electrochemical reactions within the electrolytes, (b) equivalent circuit model based on physical phenomena, and (c) simplified equivalent circuit model of EPIC actuator under biased condition.

There have been various studies on self-sensing dielectric EAP actuators featuring capacitive and resistive qualities, such as the DEA and HASEL actuator. Their sensing can be classified under electrode-resistance-based method [7], periodic high voltage (HV) method [8], and mixed-signal-based method [9]. However, the first method is sensitive to signal creep owing to compliant electrode resistance changes over time, and the second sensing method cannot sense and actuate independently. On the other hand, the third method not only senses and actuates independently but also stably senses without capacitance decrease over time. Therefore, many studies have been conducted by modifying the third method.

This paper proposes a self-sensing EPIC actuator wherein the third method is applied. Prior to designing the self-sensing system, the electrical characteristics of EPIC were investigated using electrochemical impedance spectroscopy (EIS), and the equivalent circuit model of EPIC was analyzed. Then, the mixed-signal-based self-sensing system was designed according to the electrical characteristics of EPIC. Finally, the developed self-sensing system of the EPIC actuator was successfully observed to respond to capacitance variations, and the force estimated by the self-sensing system matched well with the reference force in the loading process, with a maximum error of 0.15 N (2.8%).

II. Electrical Characteristics of EPIC Actuator

Investigation of the electrical characteristics of the EPIC actuator was first performed because the electrical characteristics largely influence the design of the sensing circuit. To analyze these characteristics, EIS was conducted.

A. Structure of EPIC Actuator

The EPIC actuator is composed of a PVC-gel film, dielectric liquid, bottom substrate, and two electrodes, as shown in Fig. 1(a). The cathode is attached to the outer layer of the PVC-gel, and the anode is attached to the upper surface of the bottom substrate. The PVC-gel is highly plasticized PVC that accumulates negative charge near the anode in an electric field. Therefore, a heterospace charge distribution (HSCD) is induced within the PVC-gel film, which then induces the zipping motion by electrostatic force, as shown in Fig. 1(b) [5]. During the zipping motion, the negative charge at the surface of the PVC-gel and positive charge at the anode form the capacitance, and the redox reaction between them is resisted by a certain activation energy. These capacitance and resistance values can largely change based on the driving voltage and applied pressure, as shown in Figs. 1(b) and 1(c).

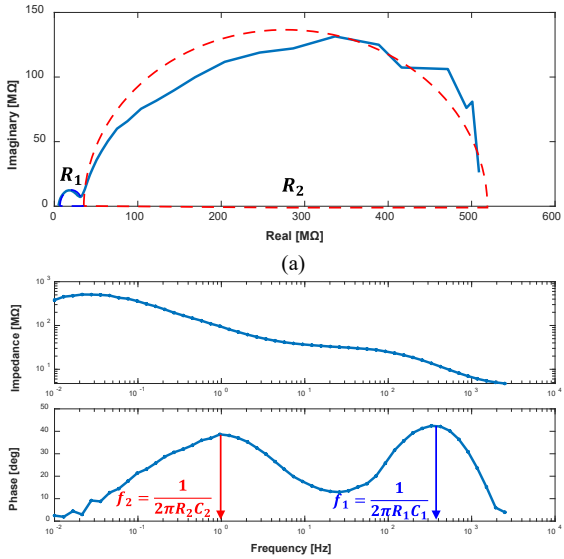


Figure 3. (a) Nyquist and (b) Bode plots of EIS results.

Table I. Values of the equivalent circuit model.

	R_1	R_2	C_1	C_2	R_s
Values	31 MΩ	478 MΩ	15.6 pF	346.8 pF	5 MΩ

B. Equivalent Circuit Model of EPIC Actuator

Fig. 2(a) shows the hypothetical electrochemical reactions in the PVC-gel. When a voltage is applied to the EPIC actuator, the charges migrate, and a double layer is formed at each electrode, resulting in the double-layer capacitance [5]. In addition, ions are the charge carriers in the electrolyte, whereas electrons are the charge carriers in the electrodes. Therefore, conduction between these two media is by electrochemical reactions, which then induce charge transfer resistance. Moreover, the electrolyte itself has a resistance because the ions within the electrolyte move under dragging force. Thus, the equivalent circuit model of the EPIC actuator based on physical phenomena is as shown in Fig. 2(b).

EIS of the EPIC actuator was next conducted under 500 V DC-bias voltage and 500 V AC voltage from 0.01 to 2500 Hz, and the parameter values were estimated from the Nyquist and Bode plots, as shown in Fig. 3. The values of each of the components are summarized in Table I. However, as shown in Fig. 3 and Table I, the parallel connection of resistor (R_2) and capacitor (C_2) dominate the overall impedance of EPIC. Therefore, the electrical circuit model of EPIC can be simplified as a parallel-connected resistor–capacitor combination under DC bias condition, as shown in Fig. 2(c).

C. Variation of Capacitance of EPIC Actuator with Respect to Contact Electrode Area

Pressing the EPIC actuator changes the contact electrode area, resulting in changes to the resistance and capacitance of

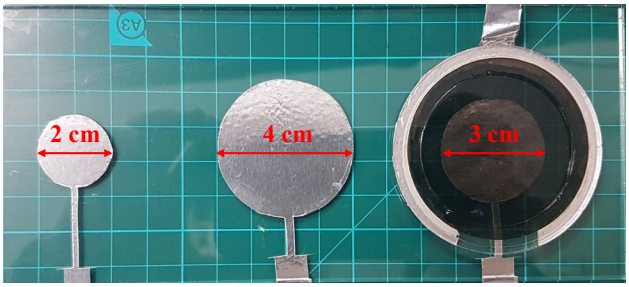


Figure 4. Sets of test electrodes with different diameters of 2cm, 3cm, and 4cm for investigating the relationships between contact electrode area and capacitance.

Table II. Relationship between capacitance and contact electrode area.

Electrode diameter (mm)	20	30	40
Area (mm^2)	314.2	706.9	1256.6
Capacitance (pF)	743	1651	2890
Capacitance/Area (pF/mm^2)	2.36	2.34	2.30

the actuator, as shown in Fig. 1. Therefore, the relationships between the contact electrode area and electrical parameter values of the EPIC actuator were investigated. EIS experiments were performed with 2 cm, 3 cm, and 4 cm diameter electrodes, as shown in Fig. 4, under 500 V DC bias voltage and 500 V AC voltage from 1 to 2500 Hz.

The results are summarized in Table II. It should be noted that the capacitance of the EPIC actuator increases proportionally with the electrode area in all cases, as is obvious from the value of the capacitance divided by the electrode area. This means that if we press the EPIC actuator, the contact electrode area decreases, and the capacitance of the EPIC actuator also decreases.

III. Circuit Design for Self-Sensing EPIC Actuator

In the previous section, the equivalent circuit model of the EPIC actuator was derived as a parallel combination of a resistor (R_a) and capacitor (C_a) through EIS measurement, as shown in Fig. 2(c). In this section, a sensing circuit to measure the capacitance change of the EPIC actuator is presented. In the developed circuit, superposition of the sinusoidal low voltage (LV) signal for sensing and HV signal for actuation is applied to the self-sensing system. Because the entire system acts as a high-pass filter, the LV high-frequency signal enables estimation of the capacitance change through the output voltage of the circuit. The schematic of the developed circuit is shown in Fig. 5. As shown in Fig. 5, the overall sensing circuit is composed of three parts: capacitance-sensitive voltage divider, high-pass filter, and peak detector.

A. Capacitance-Sensitive Voltage Divider

As shown in Fig. 5(a), an external resistor (R_m) is serially connected with the EPIC actuator. The input voltage, which is a mixed-signal of the actuation (V_a) and sensing signal (V_s), operates the EPIC actuator. The output voltage (V_0) across the external resistor can be expressed as (1), where V_a, V_s, and R_m correspond to the actuation signal, sensing signal, and external resistor, respectively. The output of the first circuit part V_0 is represented as two parts, namely V_a and V_s terms.

$$V_0 = \frac{j\omega + \omega_a}{j\omega + \omega_n} \cdot V_s + \frac{R_m}{R_a + R_m} \cdot V_a \quad (1)$$

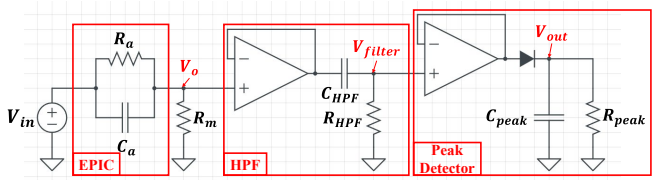


Figure 5. Structures of the capacitance-sensing circuit of EPIC actuator: (a) capacitance-sensitive voltage divider, (b) high-pass filter, and (c) peak detector.

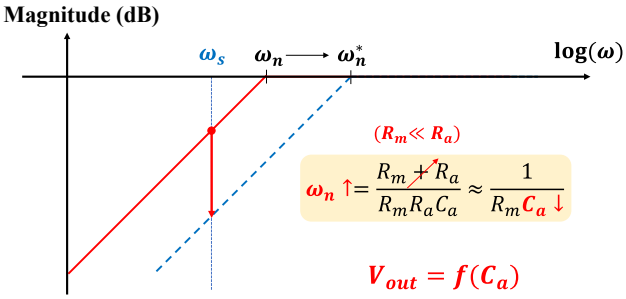


Figure 6. Bode plot of output voltage (V_{out}). C_a change leads to cut-off frequency change, so that V_{out} is represented as a function of C_a.

$$\omega_a = (R_a C_a)^{-1}, \omega_m = (R_m C_a)^{-1}, \omega_n = \omega_m + \omega_a \quad (2)$$

B. High-Pass Filter

The high-pass filter consists of a capacitor (C_{HPF}) and resistor (R_{HPF}), as shown in Fig. 5(b). This filter removes the high-voltage actuation signal term from the output voltage of the first circuit part because the actuation signal has low frequencies. Therefore, the output voltage passing through the high-pass filter (V_{filter}) can be expressed as (3).

$$V_{filter} = \frac{j\omega + \omega_a}{j\omega + \omega_n} \cdot V_s \quad (3)$$

C. Peak Detector

The developed circuit is completed by a peak detector circuit, as shown in Fig. 5(c). The choice of a small R_m compared to R_a allows approximation as (4). Therefore, the output voltage through the peak detector (V_{out}) can be expressed as (5). It should be noted that the output voltage (V_{out}) is related to the capacitance of the EPIC actuator (C_a). Since the capacitance change leads to a cut-off frequency (ω_n) change, the output voltage (V_{out}) is represented as a function of the capacitance of the EPIC actuator, as shown in Fig. 6. Therefore, the developed circuit enables self-sensing with the EPIC actuator.

$$\omega_n = \omega_m + \omega_a \approx (R_m C_a)^{-1} \quad (4)$$

$$V_{out} = |V_{filter}| \approx \left| \frac{j\omega}{j\omega + (R_m C_a)^{-1}} \cdot V_s \right| = f(C_a) \quad (5)$$

IV. Experimental Validation

A. Calibration

Using the proposed sensing circuit, the relationship between the force applied to the EPIC actuator and resulting output voltage was obtained by calibration. The experimental setup for calibration is as shown in Fig. 7(a). Using a linear motor stage, force is applied to the EPIC actuator through a flat-tip electrode of diameter 20 mm. Then, the force measured

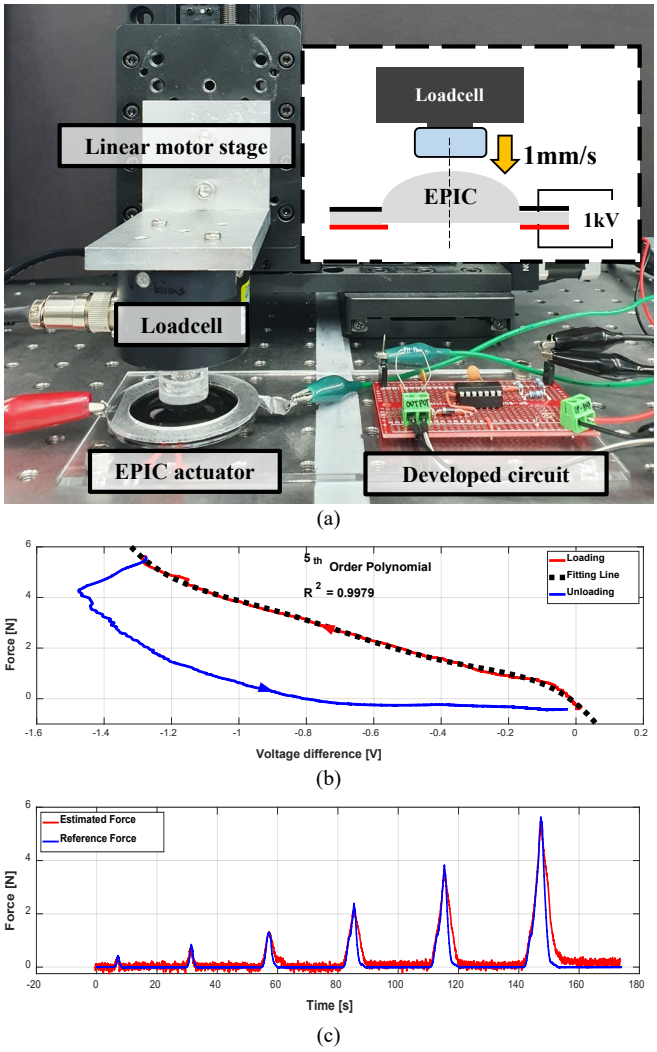


Figure 7. (a) Experimental setup for calibrating force and ΔV_{out} as well as performance evaluation. (b) Calibration results and 5th order polynomial fitting line of loading process, and (c) performance evaluation of comparison between estimated force by calibration and reference force by loadcell.

by the loadcell and output voltage measured by the sensing circuit are compared. The actuator was operated with 1 kV DC bias voltage for the actuation signal and 50 V/1000 Hz voltage for the sensing signal, and the loadcell was applied at a rate of 1 mm/s.

Fig. 7(b) shows the relationship between the applied force and measured output voltage; as seen, the output voltage follows a specific relationship with the applied force. Therefore, the self-sensing EPIC actuator was calibrated using the least-squares method with a 5th order polynomial and the loading data. However, hysteresis was observed in the graph. This hysteresis in the unloading process is interpreted to be caused by the delayed formation of a double layer that is attributed to the slow migration speed of the negative ions in the PVC-gel. Further, the creep by viscoelasticity of the PVC-gel can enhance this hysteresis error.

B. *Performance Evaluation*

To evaluate the performance of the self-sensing EPIC actuator, a repetitive force with increasing magnitude (0–5.5 N) was applied to the actuator using the experimental setup shown in Fig. 7(a). Then, the applied force was estimated using the calibration data and compared with the force measured by the loadcell.

Fig. 7(c) shows the experimental results comparing the reference and estimated forces. The developed system showed high accuracy in the loading process, with a maximum error of 0.15 N (2.8%). On the other hand, a response with time delay was observed in the unloading process. This time delay is attributed to the formation of a double layer, as mentioned in the previous section.

V. Conclusion & Future Work

This paper presents a self-sensing EPIC actuator using a capacitance sensing circuit with a mixed-signal-based self-sensing method. The self-sensing EPIC actuator showed accurate force sensing performance during the loading process, with a maximum error of 2.8%. However, there are several issues that must be addressed. First, the error in the unloading process due to the hysteresis, as shown in Fig. 7(b) and 7(c), must be addressed; to reduce the hysteresis error, future works must include further investigation of the electromechanical model of the EPIC actuator. As another solution, it is possible to reduce the hysteresis error by fabricating the EPIC actuator using a plasticizer with a smaller molecular weight [10]. In addition, the characterization of the developed system must be evaluated further, such as the effects of the loading/unloading rates and repeatability, such that the range of applications of the self-sensing EPIC actuator can be broadened.

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